THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 4: **Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $(x+2)^2 =$ 1

(a) x^2+4x+4 **(b)** x^2+2x+4 **(c)** x^2-4x+4 **(d)** x^2+4

2. $(x-4)^2 =$ 1

(a) $x^2+8x+16$ **(b)** $x^2-4x+16$ **(c)** x^2+16 **(d)** $x^2-8x+16$

3. $(x+3)(x-3) =$ 1

(a) x^2+9 **(b)** x^2-9 **(c)** x^2-6x+9 **(d)** x^2+6x-9

4. Using $(a+b)^2$, the value of 31^2 is: 1

(a) 961 **(b)** 960 **(c)** 963 **(d)** 962

5. Using a^2-b^2, the value of 49×51 is: 1

(a) 2498 **(b)** 2501 **(c)** 2499 **(d)** 2500

6. x^2-25 factorises as: 1

(a) $(x+5)(x-5)$ **(b)** $(x-5)^2$ **(c)** $(x+25)(x-1)$ **(d)** $(x+5)^2$

7. If $a=2$ and $b=5$, then $(a+b)^2 =$ 1

(a) 49 **(b)** 50 **(c)** 48 **(d)** 51

8. $(x+2)(x+3) =$ 1

(a) x^2+5x-6 **(b)** x^2+x+6 **(c)** x^2+6x+6 **(d)** x^2+5x+6

9. $(x+3)^2 =$ 1

(a) x^2-6x+9 **(b)** x^2+3x+9 **(c)** x^2+9 **(d)** x^2+6x+9

10. $(x-5)^2 =$ 1

(a) $x^2-10x+25$ **(b)** x^2+25 **(c)** $x^2+10x+25$ **(d)** $x^2-5x+25$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Expand $(x+8)^2$. 2

12. Expand $(x+8)(x-8)$. 2

13. Find 42^2 using $(a+b)^2$. 2

14. Expand $(x+10)^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Expand $(2x + 3)^2$. 3

16. Factorise $x^2 + 6x + 9$. 3

17. Evaluate 55×45 using an identity. 3

OR

Expand $(3x + 1)^2$. 3

18. Factorise $x^2 + 8x + 16$. 3

OR

Evaluate 56×44 using an identity. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Expand $(2x + y)^2$ and find its value when $x = 1$ and $y = 1$. 5

OR

Factorise $x^2 + 5x + 6$. 5

20. Expand $(3x + 2y)^2$ and find its value when $x = 1$ and $y = 1$. 5

OR

Factorise $x^2 + 7x + 12$. 5

END OF QUESTION PAPER — Total: 40 Marks

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